

Recobro Design Decisions

Design Decisions and Trade-offs

Build: Recobro. Deterministic subrogation and recovery screener. Amazon Bedrock Flows, Frankfurt (eu-central-1), Claude Sonnet 4.5 (EU inference profile).

Each entry records a decision made while building the reference implementation, the reasoning behind it, and the alternative that was considered and rejected.

1. **Orchestration: fixed workflow over an autonomous agent**
Decision: Build a fixed workflow, not an autonomous agent.
Rationale: Recovery decisions carry legal deadlines and must be repeatable and auditable on demand, not reconstructed from an agent's reasoning trace.
Rejected: An agent choosing its own sequence of investigative steps.
2. **Assessment logic: model judges criteria, code decides the band**
Decision: The model judges rubric criteria (true/false), while code decides the final band.
Rationale: The same rubric answers always produce the same band, and every band is traceable to named criteria a person can check.
Rejected: Letting the model output the band (Pursue, Investigate, or Close) directly.
3. **Arithmetic and comparisons: handled by code, not the model**
Decision: Ensure the model never performs date arithmetic or threshold comparisons.
Rationale: Dates and thresholds are exact rules with legal consequences. Models make silent arithmetic errors that are hard to catch downstream.
Rejected: Asking the model to count days remaining or compare a value to a threshold.
4. **Deadline validation: anchor check for third-party dates**
Decision: Enforce that a third-party deadline can never start before the event that caused the claim.
Rationale: Live testing showed the model correctly finding an earlier complaint date, then using it as the deadline's start, producing a deadline 156 days in the past. A claim cannot be "known" before it happens.
Rejected: Trusting the model's extracted knowledge date as the deadline anchor without a sanity check.
5. **Evidence evaluation: weak circumstantial evidence downgrades the band**
Decision: Weak circumstantial evidence downgrades to Investigate, never Pursue.
Rationale: Recommending legal pursuit on unproven signals (a missing license, a neighbour's statement) is the exact overconfidence a recovery team needs protection from.
Rejected: Treating any plausible-sounding signal as sufficient grounds to recommend pursuing recovery.
6. **PII handling: explicit Lambda nodes for masking**
Decision: PII is masked twice, as explicit Lambda nodes calling Bedrock Guardrails, not attached only to a prompt node.
Rationale: Masking needed to happen before extraction and after composition, not tied to one specific model call. Bedrock Flows only allows attaching a guardrail

directly to a prompt node.

Rejected: Attaching the guardrail only to the extraction prompt node and trusting the rest of the flow.

7. Rule configuration: versioned table over embedded prompts

Decision: The rule table is versioned configuration, not embedded in prompts or code.

Rationale: Legal can revise deadlines and thresholds without a code change, and the version number appears in every memo for traceability.

Rejected: Hardcoding deadlines or thresholds directly into the extraction or assessment prompts.

8. Model selection: Claude Sonnet 4.5 fallback

Decision: Deployed against Claude Sonnet 5 first, and fell back to Sonnet 4.5 only after confirming an account-level access gap.

Rationale: The project's own model-selection rule required trying the primary model first and switching only on a genuine failure, not a preference.

Rejected: Defaulting to Sonnet 4.5 without first attempting Sonnet 5, or switching models without confirming why the first one failed.

9. Reinsurance routing: isolated from recovery band

Decision: The reinsurance flag is a routing signal, never a factor in the recovery band.

Rationale: Aggregation for catastrophe reinsurance is a separate process from subrogation. Conflating them risks the reinsurance team acting on a signal meant for recoveries, or vice versa.

Rejected: Letting a high-value loss with the reinsurance flag active also push the band toward Pursue.